



FORD FULKERSON ALGORITHM

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You are given a directed flow network consisting of V vertices and E directed edges. Each edge has a capacity that represents the maximum amount of flow that can pass through it.

The source vertex is 0 and the sink vertex is $V-1$.

Your task is to find the maximum possible flow from the source to the sink using the Ford—Fulkerson algorithm.

Input Format

The first line contains two space-separated integers: V — number of vertices E — number of directed edges Each of the next E lines contains three space-separated integers: u — starting vertex v — ending vertex capacity — capacity of the edge

Constraints

 $2 \leq V \leq 100$ $1 \leq E \leq 1000$ $0 \leq u, v < V$ $u \neq v$ $1 \leq \text{capacity} \leq 10^6$

Output Format

Print a single integer representing the maximum flow from vertex 0 to vertex $V-1$.

Sample Input 0

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```

Sample Output 0

```
19
```

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Language

Python 3



```
1 import sys
2 from collections import deque
3
4 def solve():
5     input_data = sys.stdin.read().split()
6     if not input_data:
7         return
8
9     V = int(input_data[0])
```



```
9  def __init__(self):
10     E = int(input_data[1])
11
12     residual = [[0] * V for _ in range(V)]
13
14     adj = [set() for _ in range(V)]
15
16     idx = 2
17     for _ in range(E):
18         u = int(input_data[idx])
19         v = int(input_data[idx+1])
20         cap = int(input_data[idx+2])
21         idx += 3
22
23         residual[u][v] += cap
24
25         adj[u].add(v)
26         adj[v].add(u)
27
28     source = 0
29     sink = V - 1
30     max_flow = 0
31
32     while True:
33         parent = [-1] * V
34         parent[source] = -2
35         queue = deque([source])
```

Line: 68 Col: 1

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0	Input (stdin)		Download
	1	6 9	
	2	0 1 16	
	3	0 2 13	
	4	1 2 10	
	5	2 1 4	
	6	1 3 12	
	7	3 2 9	
	8	2 4 14	
	9	4 3 7	
	10	3 5 20	
Your Output (stdout)			